

Tracking News Consumption Across the Web: Effects on YouTube Homepage Recommendations

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Abstract

YouTube has become an important news source for many users, raising concerns about the role its recommendation algorithm may play in misinformation exposure and political polarization. Prior work has mainly studied how recommendations evolve in response to users' activity inside the platform, especially watch history. However, recommendations may also depend on off-platform browsing activity that Google observes through trackers embedded on third-party news websites, a factor that has not been considered so far.

In this work, we fill this gap by studying whether off-platform news consumption can shape YouTube homepage recommendations. We conduct a 20-day sock-puppet audit using 600 automated browser profiles that visit news articles with different political leanings or misinformation-related content before returning to YouTube to collect homepage recommendations. We run the experiment in two browser environments: a tracking-permissive environment that allows third-party tracking, and a tracking-restrictive environment that blocks trackers and third-party cookies.

Our results show that off-platform news consumption affects YouTube homepage recommendations when third-party tracking is allowed. Political news consumption shifts recommendations in the direction of the consumed news, and misinformation-related news consumption increases recommendations related to known false or misleading claims. Overall, our findings show that audits of online platforms' personalization should account for both on-platform and off-platform behavior when studying political bias and misinformation exposure.

Introduction

During the past decade, there has been a shift in the way people encounter news-related and political information online. Rather than deliberately selecting and visiting a small number of news outlets, many users now consume news as a by-product of their activity on online platforms. YouTube in particular is an important source of information. A Pew Research Center study reports that almost half of U.S. adults get news from YouTube (Pew Research Center 2025), and YouTube's Chief Product Officer states that 70% of videos watched on the platform come via its recommendation algorithm (CNET 2018).

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This raises important questions about the role of recommendation systems in shaping political and misinformation exposure. Although recommendation algorithms are designed to personalize content to users (Milano, Taddeo, and Floridi 2020), they may also contribute to selective exposure, where users repeatedly encounter content that aligns with their existing interests or beliefs (Robertson et al. 2023; Taber and Lodge 2006; Iyengar et al. 2008; Iyengar and Hahn 2009; Cardenal et al. 2019). Over time, this can support the formation of political or misinformation filter bubbles and echo chambers, where users are exposed to a small range of information and viewpoints (Pariser 2011; Bozdog and Van Den Hoven 2015; Roose 2019).

As a result, a large body of work has studied whether YouTube's recommendation system amplifies political polarization or misinformation (Chen et al. 2023; Hosseini-mardi et al. 2021; Ledwich and Zaitsev 2019; Lewis 2018). Prior studies report mixed findings. Some work finds that YouTube recommendations can pull users away from extreme content and that recommendations may skew left even in the absence of watch history (Ibrahim et al. 2023a). Other work finds that YouTube recommends ideologically aligned content to partisan users (Haroon et al. 2023). However, most previous work takes a closed-platform view, and model YouTube recommendations as a result of activity that occurs on the platform itself. These studies usually design audits that build synthetic users' watch histories with videos that have specific political or misinformation-related characteristics, and then measure how recommendations shift as those histories evolve.

In reality, YouTube's recommendation system can rely on signals beyond YouTube watch history. Google's Privacy Policy states that Google collects information about users' activity on third-party sites and apps that use Google services (e.g., Google Analytics or embedded YouTube videos) and that this information can be used to *provide personalized services, including content and recommendations across Google products* (Google 2025b,a). This suggests that browsing activity on news websites embedding Google services may affect the recommendations shown to users on YouTube. Yet, existing studies have not examined whether this off-platform browsing data contributes to political polarization or the spread of misinformation. This gap is important because Google services are widely deployed across

the web: Google Analytics alone is used by about 45% of all websites (W3Techs 2026), Google trackers appear on 72% of websites (Kieserman et al. 2025), and news websites tend to include more trackers than other types of sites (Englehardt and Narayanan 2016).

In this work, we aim to fill this gap by studying whether *off-platform news consumption influences the political and misinformation-related content in YouTube homepage recommendations*. Specifically, we investigate three research questions: (1) does off-platform browsing of political news articles shift the political bias distribution of recommended videos? (2) do off-platform visits to misinformation-related articles increase the proportion of videos related to known false or misleading claims in YouTube recommendations? and (3) do browser privacy protections that block trackers and third-party cookies reduce these recommendation shifts?

To answer these questions, we conduct a sock-puppet audit of YouTube homepage recommendations. We run 600 automated profiles across six news consumption groups: extreme-left, left, right, extreme-right, misinformation, and control. Each group visits news articles matching its assigned political or misinformation category. For example, profiles in the left group visit news articles from outlets with a left political leaning, while the control group does not visit news articles. This design allows us to test whether different types of off-platform news consumption have different effects on YouTube recommendations. In addition, for each group, we run profiles in two browser environments: a *tracking-permissive* environment, where third-party tracking is allowed, and a *tracking-restrictive* environment, where trackers and third-party cookies are blocked. This comparison allows us to test whether blocking trackers affects YouTube recommendations.

We collect recommendations over 20 days. During the first 10 days, we instruct profiles to visit news articles and then collect homepage recommendations once per day. During the next 10 days, we no longer visit news articles, but we continue collecting YouTube recommendations to measure whether any changes persist after news consumption stops. In total, we collect more than 5.3 million YouTube homepage recommendations, which we leverage to shed light on the effect of off-platform browsing activity on YouTube homepage recommendations. Our results show:

(1) *Off-platform political news consumption shifts recommendations toward the consumed news*. In the tracking-permissive environment, consumption of left-leaning news is followed by a higher share of left-leaning recommendations, while consumption of right-leaning news is followed by a higher share of right-leaning recommendations. For instance, on day 10, the left group receives 70% left-leaning recommendations and the extreme-left group receives 68%, compared with 49% for the control group. In contrast, the right group receives 58% right-leaning recommendations and the extreme-right group receives 61%, compared with 47% for the control group.

(2) *Off-platform misinformation-related news consumption increases misinformation-related recommendations*. In

the tracking-permissive environment, profiles that visit misinformation-related articles receive a higher share of YouTube recommendations related to known false or misleading claims than the control group. On day 10, the misinformation group receives 6.02% misinformation-related recommendations, compared with 0.90% for the control group, and this gap remains visible after browsing stops.

(3) *Browser privacy protections reduce these recommendation shifts*. In the tracking-restrictive environment, we do not observe the political and misinformation-related shifts seen in the tracking-permissive environment. Political recommendations remain close to the control group and do not consistently shift in the direction of the consumed news. Similarly, misinformation-related recommendations remain close to the control group.

Overall, our study shows that off-platform news consumption can shape YouTube homepage recommendations when third-party tracking of browsing activity is allowed. This means that echo chambers, filter bubbles, and misinformation exposure may not be driven only by what users do inside a platform. They may also be reinforced by the news users consume elsewhere on the web. Our results further show that browser-level privacy protections may reduce this form of content personalization. Finally, our findings show that studying YouTube recommendations only through on-platform activity gives an incomplete view. Off-platform browsing activity should also be considered when measuring how users encounter political and misinformation-related content online.

Methodology and Dataset

This section describes how we audit the effect of off-platform news consumption on YouTube homepage recommendations. We present the experimental design, describe how we select and visit news articles outside YouTube, explain how we collect and characterize homepage recommendations, introduce the browser environments, and present an overview of the dataset.

Experimental design

We conduct a systematic audit to measure whether off-platform news consumption, i.e., visits to news articles outside YouTube, affects YouTube homepage recommendations. We use automated browser instances, or *sock puppets*, to mimic users who visit news articles outside YouTube and then return to YouTube to collect homepage recommendations. We study two types of off-platform news consumption: political news consumption and misinformation-related news consumption. First, we test whether visiting news articles from outlets with different political leanings shifts the political distribution of YouTube homepage recommendations. Second, we test whether visiting articles associated with misinformation increases the share of misinformation-related recommendations.

We design the audit in three phases (Figure 1). In the (1) *setting phase*, we create sock puppets that mimic fresh browser profiles and use them to collect YouTube homepage recommendations before news consumption. We use these

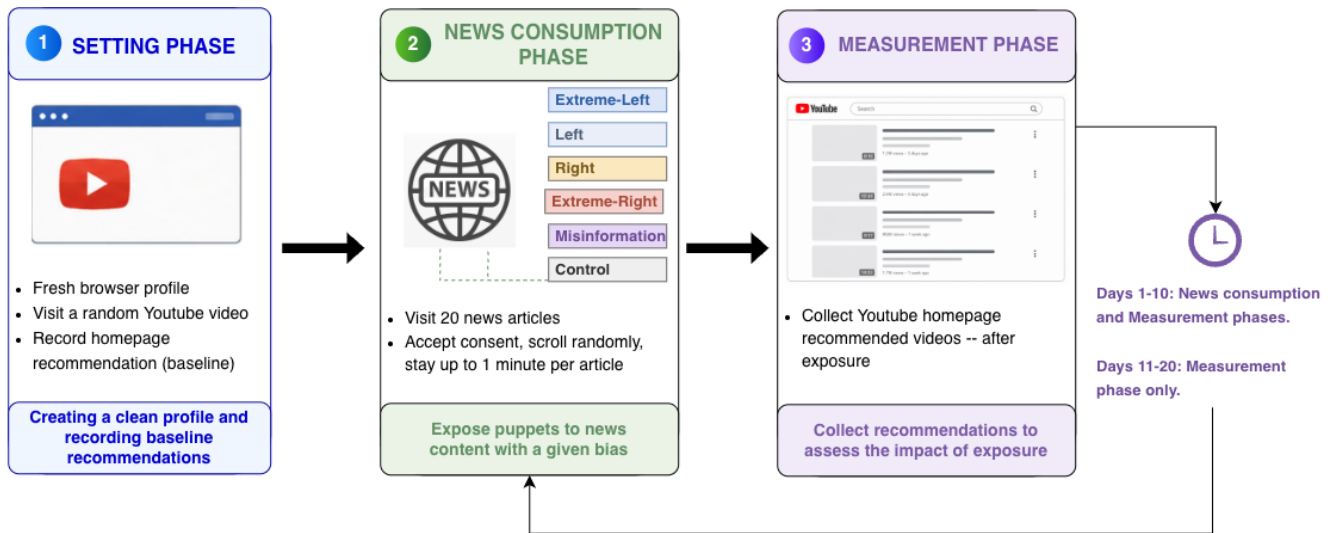


Figure 1: Diagram representing our sock puppet methodology.

recommendations as a baseline. In the (2) *news consumption phase*, we use sock puppets to visit news articles associated with different political or misinformation categories. In the (3) *measurement phase*, we collect the YouTube homepage recommendations shown after news consumption. We compare these recommendations to the baseline recommendations collected in the setting phase to measure how off-platform news browsing shifts the recommendations shown to each sock puppet. We then repeat the (2) news consumption and (3) measurement phases once per day for 10 consecutive days. This allows us to measure how YouTube recommendations change as a user repeatedly visits news articles from the same political or misinformation category over time. Finally, we continue the (3) measurement phase for 10 additional days, without off-platform news consumption. This allows us to test whether changes in recommendations persist after users stop visiting news articles. Below, we describe each phase.

Setting phase The setting phase creates a clean starting point for each sock puppet. We start each one with a fresh browser profile. A fresh profile has no browser history, cookies, local storage, or other persistent identifiers (Chromium Project 2026). This setup mimics a new user who opens a browser and visits YouTube for the first time, without an existing browsing history or persistent browser state.

At the start of the experiment, each sock puppet visits YouTube and watches the same sports-related video.¹ We include this step because we noticed that YouTube does not serve homepage recommendations to a new user until the user has watched at least one video. We choose sports content because it is not directly related to political news or misinformation. Then, we load the YouTube homepage and collect the first set of recommended videos to each sock puppet. We use these recommendations as the profile baseline. This

¹Initialization video: https://www.youtube.com/watch?v=zhEWqfP6V_w.

baseline represents what YouTube recommends before the sock puppet consumes any news articles outside YouTube.

News consumption phase During the news consumption phase, we use sock puppets to visit news articles on news websites. This phase mimics the following browsing pattern: a user may read political or misinformation-related news on the web and later open YouTube, where the homepage recommendations may reflect signals from the earlier browsing activity.

We divide sock puppets into six groups: four political groups, one misinformation group, and one control group. The political groups correspond to four political leanings: extreme-left, left, right, and extreme-right. In these groups, sock puppets visit news articles from outlets with the corresponding political leaning. In the misinformation group, sock puppets visit articles associated with false or misinformation-related claims. In the control group, sock puppets do not visit the news articles. This design allows us to compare YouTube recommendations across users who browse different types of news content.

For each sock puppet, we sample 20 articles from its corresponding group and visit them sequentially. We describe how we select the news articles visited by each group in the following Section. On each page, we attempt to accept consent banners when they appear (DuckDuckGo AutoConsent 2026). We do this to simulate users who allow cookies and tracking when visiting news sites. We then scroll through the page to simulate reading behavior and remain on the page for one minute. This allows the page to load article content, third-party scripts, ads, and tracking requests.

Measurement phase During the measurement phase, each sock puppet opens YouTube, immediately after completing the news consumption phase, to collect homepage recommendations. The homepage recommendations are the first content a user sees when they open YouTube. Measuring them allows us to capture the videos YouTube chooses to

show without giving the platform a search query or a watch action.

We collect the homepage recommendations shown to each sock puppet. We perform 20 scroll-down events to collect more recommendations from the homepage. For each recommended video, we record the title, description, video URL, and channel information.

We compare the recommendations collected during this phase to the baseline recommendations collected during the setting phase. This comparison allows us to measure whether YouTube recommendations shift after a user visits news articles outside YouTube. We repeat this measurement across 20 days to estimate how recommendations change with repeated news consumption and whether those changes persist after news consumption stops.

News article selection

We select the news articles used in the news consumption phase to represent users with different browsing habits. Our goal is to mimic users who repeatedly read news with a consistent political leaning or repeatedly visit news articles related to misinformation claims.

Political news articles. To select articles for the four political groups, we use Media Bias/Fact Check (MBFC) (Media Bias/Fact Check 2026), a journalistic organization that provides a list of news sources along with their political bias and factual reporting ratings. For each category of political bias we consider, we select 50 outlets with the corresponding MBFC label: extreme-left, left, right, and extreme-right. For each outlet, we then select 20 news article URLs from the dataset provided by Chouaki et al. (Chouaki et al. 2026). The dataset contains news article URLs published in 2022 for each of these sources. This gives us a pool of 1,000 news-article URL for each political group.

Misinformation articles. To select articles for the misinformation group, we use PolitiFact (Poynter Institute 2026), a fact-checking website that evaluates political and public-interest claims. We use PolitiFact because it links fact-checked claims to the articles or sources in which those claims appear. We collect URLs associated with claims that PolitiFact labels as *false* or *pants-fire* between 2020 and 2025. We then keep only URLs from domains listed by Media Bias/Fact Check to ensure that we only include news outlets. In total, this gives us 659 article URLs from 127 distinct source domains, linked to 568 unique PolitiFact fact-checks.

All article pools reported above are obtained after applying two eligibility filters. First, the article URL must be reachable. We check this by sending an HTTP request to the URL and recording the response status code. We do not include URLs that return 4XX or 5XX responses. Second, the article must be in a news domain that includes at least one Google tracker. We impose this requirement because our goal is to study whether off-platform news consumption can influence YouTube news recommendations. Hence, we focus on article visits that Google services can plausibly observe. If a news outlet does not include any Google tracker, user activity on that outlet is unlikely to be recorded by Google services.

To check for Google trackers, we crawl the landing page of each outlet and record the network traffic. We then check whether the page reaches Google-owned tracking or advertising domains. We consider an outlet to include a Google tracker if its landing page contacts at least one of the following domains: google-analytics.com, googletagmanager.com, doubleclick.net, googlesyndication.com, googleadservices.com, or googletagservices.com.

Browser environments

We run the audit in two browser environments. The first is a *tracking-permissive* environment. In this configuration, we use Google Chrome with default settings (Google 2026). Google Chrome allows third-party cookies and does not block third-party tracking requests (Google Chrome Help 2026). Therefore, Google trackers are more likely to record sock puppets’ visits to news articles.

The second is a *tracking-restrictive* environment. In this configuration, we use the Brave browser with Brave Shields enabled (Brave 2026a). Brave Shields block third-party cookies and trackers (Brave 2026b). Therefore, Google trackers are less likely to record sock puppets’ visits to news articles through third-party tracking mechanisms. This setting represents users who browse the web with browser-level privacy protections.

We use these two environments to evaluate whether off-platform news consumption affects YouTube recommendations differently depending on the browser environment. This comparison allows us to measure whether a privacy-focused browser reduces potential shifts in political and misinformation-related recommendations.

Dataset

We deploy 600 sock-puppet profiles to collect YouTube homepage recommendations. We run 100 sock puppets for each of the six news-articles consumption groups: extreme-left, left, right, extreme-right, misinformation, and control. For each group, we run 50 profiles in the tracking-permissive environment and 50 profiles in the tracking-restrictive environment. This gives us a total of 300 profiles in Google Chrome and 300 profiles in Brave.

We run all sock puppets over the same 20-day period, from 2026-03-11 to 2026-03-30. Each sock puppet first runs the setting phase once to collect its baseline recommendations. Then, during the first 10 days, each sock puppet runs the news consumption phase followed by the measurement phase once per day. This means that it visits news articles and then collects YouTube homepage recommendations each day. During the last 10 days, each sock puppet only runs the measurement phase. This means that it continues to collect homepage recommendations, but does not visit additional news articles.

In total, we collect 5,399,466 recommended videos, with an average of 429 recommendations per sock puppet per day. Table 1 presents the number of videos per browser environment and news consumption group.

Table 1: Number of sock-puppet profiles and YouTube homepage recommendations collected for each experimental group.

Condition	# Sock-puppet Profiles	# Videos
<i>Tracking-permissive environment</i>		
<i>Extreme-left</i>	50	457,180
<i>Left</i>	50	429,364
<i>Right</i>	50	436,366
<i>Extreme-right</i>	50	455,467
<i>Misinformation</i>	50	444,840
<i>Control</i>	50	441,001
<i>Tracking-restrictive environment</i>		
<i>Extreme-left</i>	50	454,069
<i>Left</i>	50	445,517
<i>Right</i>	50	456,322
<i>Extreme-right</i>	50	457,378
<i>Misinformation</i>	50	459,496
<i>Control</i>	50	462,366
Total	600	5,399,466

Characterizing recommended videos

After collecting YouTube homepage recommendations, we characterize each recommended video along two dimensions. First, we assign a political-leaning label to estimate the ideology of the recommendation. Second, we compute a misinformation score to estimate how close the video is to known false or misleading claims. These two measures allow us to compare how different types of off-platform news consumption change the political and misinformation-related content shown to users on YouTube.

Political leaning We assign a political-leaning label to each recommended video using a fine-tuned BERT classifier proposed by AlDahoul et al. (AlDahoul, Rahwan, and Zaki 2024). The classifier predicts the political leaning of a YouTube video from its title. For each title, it outputs a probability distribution over six labels: *far-left*, *left*, *center*, *anti-woke*, *right*, and *far-right*, then assigns each video the label with the highest predicted probability. AlDahoul et al. (AlDahoul, Rahwan, and Zaki 2024) report that this BERT classifier achieves 75% accuracy and 77% F1 score. The classifier is publicly released by the authors under the MIT License (comnetsAD 2024).

We apply the classifier to videos with English titles and filter out videos with non-English titles. Overall, we consider 4,352,821 videos (80.6% of the dataset). Among these, 21.9% are classified as *far-left*, 26.9% as *left*, 10.7% as *center*, 19.3% as *anti-woke*, 7.1% as *right*, and 14.2% as *far-right*.

Misinformation score Similar to prior work (Ibrahim et al. 2023a), we compute a misinformation score by comparing each recommended video title to a reference set of false or misleading claims from PolitiFact. The score measures how semantically close a recommended video title is to known false or misleading claims. A higher score means

that the YouTube title is closer to a known false or misleading claim, and therefore more likely to be related to misinformation.

We collect all fact-checked claims that PolitiFact labels as *false* or *pants-fire* between January 2020 and March 2026. This gives us 7,992 PolitiFact fact-check headlines. These headlines represent claims that fact-checkers identified as false or misleading. We then encode both YouTube video titles and PolitiFact headlines using the `all-mpnet-base-v2` SentenceTransformer model (Sentence Transformers 2026). For each recommended video, we compute the cosine similarity between its title embedding and the embeddings of all PolitiFact headlines. We keep the highest cosine similarity score as the video’s misinformation score.

Results and Evaluation

This section presents the results of our audit. We first analyze political recommendation patterns, then examine misinformation-related recommendations, and finally compare the tracking-permissive and tracking-restrictive browser environments.

Political recommendation patterns

We first analyze the political composition of YouTube homepage recommendations. We consider sock puppets running in the tracking-permissive environment and assigned to the four political news consumption groups: extreme-left, left, right, and extreme-right. These sock puppets visit news articles from outlets with the corresponding political leaning before returning to YouTube. We compare their homepage recommendations with those collected from the control group, which does not visit news articles.

We include only videos for which we assign a political-bias label, and inspired by prior work (Haroon et al. 2023), we map each political-bias label to a numeric score on a left-to-right scale. We map *far-left* to -1 , *left* to -0.5 , *center* to 0 , *anti-woke* and *right* to 0.5 , and *far-right* to 1 . Then, for each sock puppet and day, we compute the *weighted average political slant* across its recommendations.

We use a weighted average because videos do not have the same visibility on the homepage. Videos near the top of the feed are more visible than videos shown after several scrolls, so we assign them higher weights when computing the average political slant. YouTube displays homepage recommendations in rows of four videos. We therefore assign the same weight to videos in the same row. We then apply a logarithmic discount across rows, following prior work on ranked-list evaluation, where items shown lower in a list receive less weight (Järvelin and Kekäläinen 2002; Wang et al. 2013).

Let R_v be the row number of video v . We define the weight of video v as in Equation 1. For example, videos in row 1 receive weight 1.00, videos in row 2 receive weight 0.63, videos in row 5 receive weight 0.39, and videos in row 10 receive weight 0.29.

$$w(v) = \frac{1}{\log_2(R_v + 1)} \quad (1)$$

We then average scores across all sock puppets in the same news consumption group. The resulting group-level score summarizes the political slant of recommendations shown to that group on that day. The score ranges from -1 to 1 : negative values indicate that recommendations lean more to the left, positive values indicate that recommendations lean more to the right, and values close to 0 reflect a more politically balanced distribution. Figure 2a presents the day-by-day evolution of the average political slant score across the five groups. The figure shows three main patterns.

First, the control group remains relatively stable throughout the experiment. Its score ranges from -0.18 to 0.01 . This suggests that YouTube homepage recommendations are slightly left-leaning by default, and, without news visits, their political distribution does not shift strongly over time.

Second, the left and extreme-left groups shift toward more negative scores. Starting from day 3, both groups remain below the control group, meaning that their recommendations are more left-leaning. For example, on day 10, the left group has a political slant score of -0.36 and the extreme-left group has a score of -0.31 , compared with -0.03 for the control group. This pattern persists during the measurement-only period. On day 20, the left group remains at -0.38 and the extreme-left group at -0.26 , compared with -0.05 for the control group.

Third, the right and extreme-right groups shift in the opposite direction. The extreme-right group is above the control group from day 2 onward, and the right group is above the control group on most days after news consumption begins. For example, on day 10, the right group has a political slant score of 0.17 and the extreme-right group has a score of 0.20 , compared with -0.03 for the control group. This shift also persists after news consumption stops. On day 20, the right group has a score of 0.03 and the extreme-right group has a score of 0.15 , while the control group remains at -0.05 .

Moreover, Figure 3 in the Appendix presents the weighted political distribution for each group and day. This distribution confirms the trends observed with the political slant score. The left and extreme-left groups show a larger share of left-leaning recommendations, while the right and extreme-right groups show a larger share of right-leaning recommendations.

We further test whether these shifts are statistically different from the shifts observed in the control group. Specifically, we test whether each political news consumption group (i.e., extreme-left, left, right, and extreme-right) differs from the control group in two periods: the news consumption and measurement period (days 1-10), and the measurement-only period (days 11-20), with respect to the baseline (day 0). For each sock puppet, we compute its average political slant score during each period and subtract its own baseline score. This gives one shift-from-baseline value per sock puppet for each period. We then compare the distribution of shift-from-baseline values in each political news consumption group with the corresponding distribution in the control group using a Mann-Whitney U test. We present more details in the Appendix.

The tests show that the left, extreme-left, and right groups

significantly differ from the control group during both periods ($p < 0.001$). The extreme-right group differs from the control group during the news consumption period ($p < 0.001$), but not during the measurement-only period ($p = 0.$).

Overall, these results show that off-platform political news consumption shifts YouTube homepage recommendations in the direction of the consumed news. The effect is visible for both left-leaning and right-leaning news consumption, although the persistence of the effect differs across groups.

Misinformation recommendation patterns

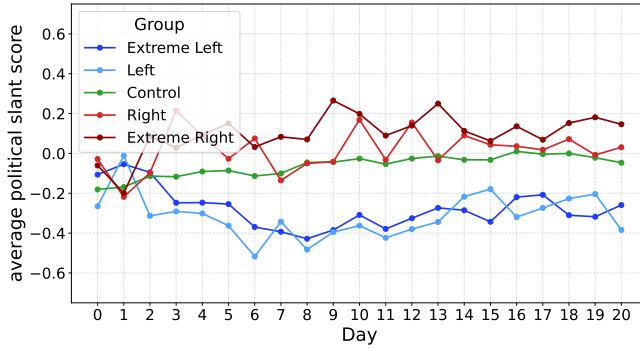
We next examine how off-platform consumption of misinformation-related articles affects YouTube homepage recommendations. We focus on sock puppets assigned to the misinformation group and running in the tracking-permissive environment. These profiles visit news articles linked to claims that PolitiFact labels as *false* or *pants-fire* before returning to YouTube. We compare their homepage recommendations with those collected from the control group, which does not visit news articles.

For each experiment day, we compute the average proportion of recommendations related to misinformation claims. We use the same row-based weighting formula described in Equation 1, so that videos shown higher on the homepage contribute more to the overall proportion. For each recommended video, we use the misinformation score we described earlier. This score, between 0 and 1 , measures the cosine similarity between the video title and a reference set of false or misleading claims from PolitiFact. Higher values indicate that a recommended video title is semantically closer to a known false or misleading claim. We classify a video as misinformation-related when its cosine similarity score is higher than a threshold τ .

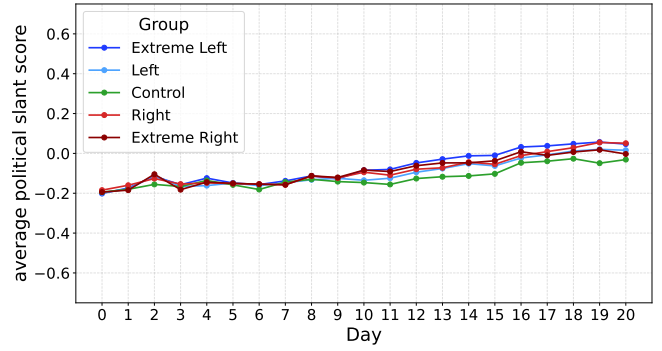
Table 2a reports the day-by-day proportion of misinformation-related recommendations for the misinformation and control groups in the tracking-permissive environment. Similar to previous work (Ibrahim et al. 2023a), we use a threshold $\tau = 0.80$ to identify misinformation-related recommendations.

The table shows that consumption of misinformation-related articles is followed by an increase in misinformation-related recommendations. The misinformation group starts close to the control group at baseline, with 0.51% of recommendations classified as misinformation-related, compared with 0.72% for the control group. The difference appears during the period in which we run both the news consumption phase and the measurement phase. From day 4 onward, the misinformation group remains consistently above the control group. For example, on day 10, 6.02% of recommendations in the misinformation group are classified as misinformation-related, compared with 0.90% in the control group.

This effect also persists after news consumption stops. During the measurement-only period, the misinformation group remains above the control group on every day. The average proportion of misinformation-related recommenda-



(a) Tracking-permissive environment



(b) Tracking-restrictive environment

Figure 2: Day-by-day evolution of the weighted average political slant score in the two browser environments. Day 0 is the baseline collected during the setting phase. Days 1–10 correspond to the period with both news consumption and measurement phases. Days 11–20 correspond to the measurement-only period.

tions is 5.53% for the misinformation group during days 11–20, compared with 1.15% for the control group.

We also test whether the increase in misinformation-related recommendations is statistically different from the shift observed in the control group. Similar to political shifts, we test whether the misinformation group differs from the control group in two periods: the news consumption and measurement period, covering days 1–10, and the measurement-only period, covering days 11–20, with respect to the baseline on day 0. For each sock puppet, we compute its average proportion of misinformation-related recommendations during each period and subtract its own baseline proportion. This gives one shift-from-baseline value per sock puppet for each period. We then compare the distribution of shift-from-baseline values in the misinformation group with the corresponding distribution in the control group using a Mann-Whitney U test. We present more details in the Appendix.

The tests show that the misinformation group differs from the control group during both the news consumption and measurement period and the measurement-only period ($p < 0.001$).

Effect of browser privacy protections

We finally examine whether browser-level privacy protections change the effect of off-platform news consumption on YouTube homepage recommendations. We compare the tracking-permissive and tracking-restrictive environments across both political- and misinformation-related recommendations. The tracking-permissive environment allows third-party cookies and tracking requests, while the tracking-restrictive environment blocks third-party trackers and cookies.

For political recommendations, we compare the weighted average political slant score across the tracking-permissive and tracking-restrictive environments in Figure 2. In the tracking-permissive environment, off-platform political news consumption produces clear shifts in the direction of the consumed news, as discussed in previous sections. In

contrast, the tracking-restrictive environment shows much weaker shifts. During the news consumption and measurement period, all political groups remain close to the control group, with average differences ranging from 0.01 to 0.02 points. During the measurement-only period, the differences remain smaller than in the tracking-permissive environment and do not follow the expected left-right direction. All political groups are slightly above the control group, with average differences ranging from 0.04 to 0.08 points.

We observe a similar pattern for misinformation-related recommendations. Comparing Tables 2a and 2b shows that the increase in misinformation-related recommendations occurs only in the tracking-permissive environment. In the tracking-restrictive environment, the misinformation group is slightly below the control group during the period in which we run both the news consumption and measurement phases, with an average difference of -0.14% , and only 0.25% higher during the measurement-only period.

Overall, these results indicate that browser-level privacy protections reduce the effect of off-platform news consumption on YouTube homepage recommendations.

Related work

Several studies have explored the dynamics of news, misinformation, political content, and extremist content on the web. Prior work has studied these questions across multiple platforms, including news media websites (Flaxman, Goel, and Rao 2016a; Guess, Nyhan, and Reifler 2020; Scharkow et al. 2020; Ognyanova et al. 2020; Flaxman, Goel, and Rao 2016b), Google News (Le et al. 2019), Twitter (Grinberg et al. 2019; Mosleh et al. 2021; Eady et al. 2019, 2023), Facebook (Bakshy, Messing, and Adamic 2015; Hargreaves et al. 2020; Amieur et al. 2025; Chouaki et al. 2024; Guess, Nagler, and Tucker 2019; Levy 2021; Coelho et al. 2023), and YouTube. We focus on studies that examine YouTube, as they are closest to our work.

We divide YouTube-focused studies into three categories based on their methodology (Chandio, Dar, and Nithyanand 2024). The first category uses web-browsing histories to

(a) Tracking-permissive environment																					
Group	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
Misinformation	0.51	1.98	2.05	1.27	4.74	4.31	3.49	4.73	5.52	7.48	6.02	5.17	5.15	4.05	5.09	5.97	6.46	4.32	7.27	6.01	5.79
Control	0.72	0.85	1.24	1.39	1.57	1.26	1.14	0.97	1.36	1.66	0.90	0.94	0.99	1.39	1.10	0.89	1.10	1.14	1.87	1.35	0.72

(b) Tracking-restrictive environment																					
Group	D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
Misinformation	0.69	0.89	0.72	1.22	1.19	0.65	1.23	1.08	0.87	0.85	0.94	1.17	0.99	1.23	1.64	0.94	0.85	1.10	0.76	1.59	1.30
Control	0.60	1.08	0.79	1.39	0.86	1.41	1.17	1.00	0.74	1.64	0.94	0.77	0.70	0.72	0.48	0.73	0.76	0.96	0.98	1.62	1.34

Table 2: Day-by-day evolution of the average proportion (%) of misinformation-related YouTube homepage recommendations. We classify a video as misinformation-related if the cosine similarity between the video and a set of known false articles from Politifact is greater than $\tau = 0.80$. Day 0 is the baseline we collect in the setting phase. Days 1–10 correspond to the period with both the news consumption and the measurement phases, and days 11–20 correspond to the measurement-only period.

observe the YouTube videos that real users visit. The second category uses crawlers or API-based collection to study YouTube videos, channels, comments, and engagement patterns. The third category uses sock-puppet audits, where researchers create controlled user profiles and measure the recommendations presented to them.

Web-browsing history based studies

Web-browsing history based studies construct real users’ news exposure on YouTube by observing the URLs that the users visit. These URLs can be collected through monitoring tools, browser extensions, or data donations. The researchers then classify the visited URLs to identify news, misinformation, political content, extremist content, or other types of online exposure.

Several studies use this approach to study YouTube. Hosseinmardi et al. (Hosseinmardi et al. 2021) examine the consumption of radical content on YouTube using browsing histories. They show that users who consume radical content on YouTube also tend to consume similar political content elsewhere on the web. Chen et al. (Chen et al. 2023) study alternative and extremist YouTube channels using browsing data. They show that exposure to these channels is concentrated among users with pre-existing resentful attitudes, and that these users often reach such channels through subscriptions or links from outside YouTube.

Platform-data and crawler-based studies

A second line of work collects YouTube videos through the YouTube API or using crawlers, to analyze political communication, misinformation, or extremist content. For example, Ribeiro et al. (Ribeiro et al. 2020) study radicalization pathways on YouTube by collecting channel information, videos, and comments using crawlers. Hussein et al. (Hussein, Juneja, and Mitra 2020) audit misinformation in YouTube search results and Up-Next recommendations by building a crawler that collect videos returned by YouTube search for a set of search queries. More recent works use the YouTube API to collect data from selected channels and study political communication and media discourse during elections (Violot, Sosnovik, and Humbert 2026; Sosnovik, Violot, and Humbert 2026).

Sock-puppet based audits

A third line of work uses sock-puppet audits to study YouTube recommendations. Sock-puppet audits use automated accounts or browser profiles to simulate users under controlled conditions. Researchers assign different behaviors to different profiles, such as watching videos from a given ideology or interacting with specific types of content. They then compare the recommendations shown to these profiles.

Several studies use sock-puppet audits to study YouTube recommendations. Haroon et al. (Haroon et al. 2023) use automated browser instances to audit whether YouTube recommends ideologically congenial, extreme, and problematic videos. Ibrahim et al. (Ibrahim et al. 2023b) use political personas to measure whether YouTube recommendations lean left or right in the U.S. political context. Srba et al. (Srba et al. 2023) uses sock puppets to explore misinformation filter bubbles on YouTube.

Our study falls into this category, but introduces a different way of constructing sock-puppet profiles. Most prior YouTube sock-puppet audits create these profiles through on-platform activity. They build profiles by making the sock puppets watch YouTube videos associated with specific ideologies, topics, or misinformation narratives, and then measure the recommendations generated for those profiles. In contrast, we construct sock-puppet profiles through off-platform browsing activity. We assign profiles to visit news articles on external websites before measuring their YouTube homepage recommendations. This design allows us to test how news consumption outside YouTube can affect recommendations inside YouTube.

Concluding Discussion

In this work, we studied whether off-platform news consumption can affect YouTube homepage recommendations. Prior audits of YouTube recommendations study personalization as a response to activity that happens inside YouTube. We take a different approach and test whether browsing activity outside YouTube can also affect the homepage recommendations. To do this, we develop a sock-puppet based methodology in which automated browser profiles visit political or misinformation-related news articles on external websites, then return to YouTube to col-

lect homepage recommendations. We compare these profiles with a control group that does not visit news articles.

We assign profiles to different news consumption groups: extreme-left, left, right, extreme-right, misinformation, and control. We then measure how the political and misinformation-related composition of YouTube homepage recommendations changes over time. We also run the audit in two browser environments: a tracking-permissive environment, where third-party tracking is allowed, and a tracking-restrictive environment, where trackers and third-party cookies are blocked.

Limitations

Our study has two main limitations. First, we use a BERT-based classifier to predict the political leaning of YouTube videos from their titles. Although this classifier reports strong performance, with 75% accuracy and 77% F1 score (AlDahoul, Rahwan, and Zaki 2024), it may mislabel some videos and therefore affect the estimated political slant of recommendations.

Second, the misinformation score is based on semantic similarity between YouTube video titles and PolitiFact claims. This allows us to measure whether recommendations are close to known false or misleading claims at scale, but it does not determine whether a video promotes misinformation. A title can be similar to a fact-checked claim because the video promotes the claim, criticizes it, or debunks it. Another limitation is that we use only video titles. The prior work that introduced this methodology compares fact-checked claims with both video titles and transcripts (Ibrahim et al. 2023a). Therefore, if a misinformation claim appears only in the transcript and not in the title, our score will not capture it. Unfortunately, collecting transcripts for our full dataset of more than 5.3 million recommended videos is challenging due to rate limitations.

To the best of our knowledge, no existing methodologies fully address these limitations. They are common to large-scale studies of political and misinformation-related patterns online, and they remain an open challenge for the research community. Moreover, our analysis relies on aggregate trends across a large number of recommendations and sock-puppet profiles. Therefore, isolated classification errors are less likely to drive the overall patterns we report.

Implications of results

We highlight four noteworthy aspects of our findings:

Cross-site tracking and information exposure. Our findings suggest a broader mechanism through which political bubbles and misinformation exposure may form. A user who reads political or misinformation-related content on the web may later receive YouTube recommendations that align with that off-platform activity. This means that selective exposure may not be driven only by what users watch, search for, or click inside a platform. It may also be reinforced by the content users consume elsewhere on the web.

Browser privacy protections. Our results also highlight the role of browser-level privacy protections. We do not

observe the political and misinformation-related shifts in the tracking-restrictive environment. This suggests that blocking trackers and third-party cookies, using privacy-enhancing tools, may reduce content personalization.

Auditing online platforms. Most prior audits of recommender systems on online platforms model personalization as a result of activity that happens inside the platform, such as watch history. Our findings show that this view is incomplete. Recommendations may also reflect activity that occurs outside the platform, when third-party tracking is allowed. Future audits should therefore consider both on-platform and off-platform behavior when studying personalization, political bias, and online misinformation exposure. To support future work in this direction, we make our experimental framework publicly available (Anonymous Github Repository 2026).

Regulation and user control. Finally, our findings are relevant to ongoing regulatory discussions about recommender-system transparency and user control. For example, the EU Digital Services Act requires online platforms to explain the main parameters used in their recommender systems and the options users have to modify or influence those parameters (European Union 2022). Our results suggest that such transparency should not be limited to signals collected inside a platform, but should also disclose whether external browsing signals are used for personalization. Similarly, user controls should allow people to limit personalization based on activity collected across the web.

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AAAI ICWSM Paper Checklist

1. For most authors...

- (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? **Yes. We believe that the study advances understanding of how personalization on online platforms works without violating privacy norms. We rely on controlled sock-puppet profiles and public news articles, and the goal is to audit platform behavior. We do not rely on or use real-user data.**
- (b) Do your main claims in the abstract and introduction accurately reflect the paper's contributions and scope? **Yes.**
- (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? **Yes. This information is mainly presented in the "Methodology and Dataset" section, and is also discussed in the "Introduction" and "Concluding Discussion" sections.**
- (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? **NA. Our research does not involve users. Hence, no populations-specific distributions).**
- (e) Did you describe the limitations of your work? **Yes, we include a "Limitations" subsection in the "Concluding Discussion" section of the paper.**
- (f) Did you discuss any potential negative societal impacts of your work? **No, we do not expect the work to have any negative societal impact.**
- (g) Did you discuss any potential misuse of your work? **No, we do not believe that the work has the potential to be misused.**
- (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? **No. We do not believe the study creates negative outcomes that require additional mitigation.**
- (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? **Yes.**

2. Additionally, if your study involves hypotheses testing...

- (a) Did you clearly state the assumptions underlying all theoretical results? **NA. Our study is an empirical audit and does not present theoretical results.**
- (b) Have you provided justifications for all theoretical results? **N/A. Our study does not present theoretical results.**
- (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? **NA.**
- (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? **NA.**
- (e) Did you address potential biases or limitations in your theoretical framework? **NA. We do not propose a theoretical framework. We discuss limitations of our empirical methodology, including political-label classification and misinformation scoring in the "Limitations" subsection.**
- (f) Have you related your theoretical results to the existing literature in social science? **NA. We do not have theoretical results.**
- (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? **NA. We do not have theoretical results. However, we do discuss implications of our results in the "Concluding Discussion" section.**

3. Additionally, if you are including theoretical proofs...

- (a) Did you state the full set of assumptions of all theoretical results? **NA. No theoretical proofs.**
- (b) Did you include complete proofs of all theoretical results? **NA. No theoretical proofs.**

4. Additionally, if you ran machine learning experiments...

- (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **Yes. We provide an anonymized repository, cited in the "Concluding Discussion," that includes the pre-trained model, the code needed to run the model, our experimental framework, analysis scripts, and instructions to reproduce the experiments. We do not include the collected dataset in the repository because it is very large.**

Upon acceptance, we will state in the paper that the dataset can be shared upon request.

- (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? No. We do not train a new model in this work. We use a pre-trained classifier provided by prior work, which reports the training details and releases the model for classification.
 - (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? NA
 - (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? No. We did not include this information because we do not train a new model. We use a pre-trained classifier, and running our classification and analysis pipeline does not require high compute resources. The pipeline can be run on a standard computer, so we do not consider compute details central to the reproducibility of this work.
 - (e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? No. We do not propose new classification methods. Our analysis relies on established methods from prior work. We report accuracy / F1 score. We describe how these methods are applied in our setting and use them to support the empirical claims made in the paper.
 - (f) Do you discuss what is “the cost” of misclassification and fault (in)tolerance? Yes. We discuss this in the limitations section. We explain that political-leaning classification and misinformation-related scoring are imperfect and may misclassify some videos. However, our analysis relies on aggregate trends across a large number of recommendations and sock-puppet profiles, which makes isolated classification errors less likely to drive the overall patterns we report.
5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...
- (a) If your work uses existing assets, did you cite the creators? Yes
 - (b) Did you mention the license of the assets? Yes.
 - (c) Did you include any new assets in the supplemental material or as a URL? No. We just cite the anonymized github repository in the “Concluding Discussion” section.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? NA. The data was not obtained from people.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? NA. There are no PII’s involved in the study.
 - (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see FORCE11 (2020))? NA. The collected dataset is very large (hundreds of GB), so we do not include it

in the repository. Instead, we release the experimental framework and analysis scripts needed to reproduce the data collection and analyses. Upon acceptance, we will state in the paper that the dataset can be shared upon request.

- (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al. (2021))? NA.

6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity**...

- (a) Did you include the full text of instructions given to participants and screenshots? NA.
- (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? NA
- (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? NA
- (d) Did you discuss how data is stored, shared, and de-identified? NA

References

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Statistical test for political recommendation shift

We test whether the political recommendation shifts observed for the extreme-left, left, right, and extreme-right groups are statistically different from the shifts observed in the control group. To do so, we compare the distribution of these shifts at the sock-puppet level using Mann-Whitney U test.

Let $S_{i,d}$ be the weighted average political slant score of sock puppet i on day d . Day 0 corresponds to the baseline collected during the setting phase, before any news consumption. For each sock puppet, we compute its average political slant score during the news consumption and measurement period, covering days 1-10, as shown in Equation 2. We also compute its average political slant score during the measurement-only period, covering days 11-20, as shown in Equation 3. We then subtract the sock puppet's own baseline score from each period average. This gives one shift-from-baseline value per sock puppet for each period, as shown in Equations 4 and 5.

$$\bar{S}_{i,1:10} = \frac{1}{10} \sum_{d=1}^{10} S_{i,d} \quad (2)$$

$$\bar{S}_{i,11:20} = \frac{1}{10} \sum_{d=11}^{20} S_{i,d} \quad (3)$$

$$\Delta_i^{1:10} = \bar{S}_{i,1:10} - S_{i,0} \quad (4)$$

$$\Delta_i^{11:20} = \bar{S}_{i,11:20} - S_{i,0} \quad (5)$$

For each political news consumption group $g \in \{\text{extreme-left}, \text{left}, \text{right}, \text{extreme-right}\}$, we compare the distribution of shift-from-baseline values in group g with the corresponding distribution in the control group. We run this comparison separately for the news consumption and measurement period and for the measurement-only period. For each comparison, the null hypothesis is that the shift-from-baseline values in the treatment group and the control group come from the same distribution. The alternative hypothesis is that the two distributions differ:

We use a Mann-Whitney U test for all comparisons. This test compares two independent groups and does not assume normal distributions. We use a significance threshold of $\alpha = 0.05$. Table 3 reports the results

The results show that the extreme-left, left, and right groups differ from the control group in both periods. The extreme-right group differs from the control group during the news consumption and measurement period, but not during the measurement-only period.

Statistical test for misinformation recommendation shifts

We test whether the misinformation recommendation shifts observed for the misinformation group are statistically different from the shifts observed in the control group. We perform the test at the sock-puppet level.

Let $M_{i,d}$ be the weighted proportion of misinformation-related recommendations shown to sock puppet i on day d . Day 0 corresponds to the baseline collected during the setting phase, before any news consumption. For each sock puppet, we compute its average proportion of misinformation-related recommendations during the news consumption and measurement period (Equation 6), and during the measurement-only period (Equation 7). We then subtract the sock puppet's own baseline value from each period average (Equations 8 and 9). This gives one shift-from-baseline value per sock puppet for each period.

$$\bar{M}_{i,1:10} = \frac{1}{10} \sum_{d=1}^{10} M_{i,d} \quad (6)$$

$$\bar{M}_{i,11:20} = \frac{1}{10} \sum_{d=11}^{20} M_{i,d} \quad (7)$$

$$\Delta_i^{1:10} = \bar{M}_{i,1:10} - M_{i,0} \quad (8)$$

$$\Delta_i^{11:20} = \bar{M}_{i,11:20} - M_{i,0} \quad (9)$$

We compare the distribution of shift-from-baseline values in the misinformation group with the corresponding distribution in the control group. We run this comparison separately for the news consumption and measurement period and for the measurement-only period. For each comparison, the null hypothesis is that the shift-from-baseline values in the misinformation group and the control group come from the same distribution. The alternative hypothesis is that the two distributions differ.

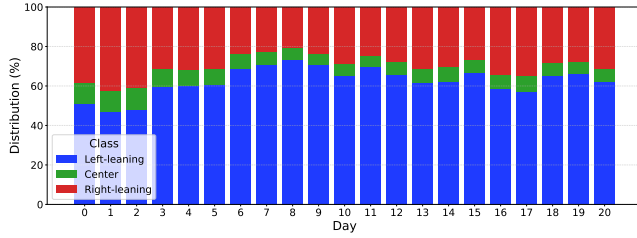
We use a Mann-Whitney U test using a significance threshold of $\alpha = 0.05$. Table 4 reports the results.

Group	Period	p -value
Extreme-left	Days 1–10	3.29×10^{-17}
Extreme-left	Days 11–20	1.31×10^{-19}
Left	Days 1–10	8.32×10^{-19}
Left	Days 11–20	2.28×10^{-17}
Right	Days 1–10	2.36×10^{-6}
Right	Days 11–20	1.69×10^{-6}
Extreme-right	Days 1–10	5.22×10^{-5}
Extreme-right	Days 11–20	0.083

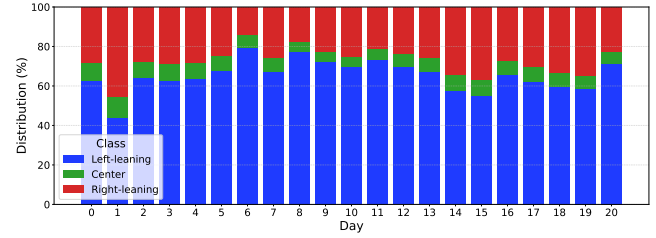
Table 3: Mann–Whitney U tests comparing shift-from-baseline political slant scores between each political news consumption group and the control group. The test is run separately for the news consumption and measurement period, covering days 1–10, and the measurement-only period, covering days 11–20.

Table 4: Mann–Whitney U tests comparing shift-from-baseline misinformation-related recommendation proportions between the misinformation group and the control group. The test is run separately for the news consumption and measurement period, covering days 1–10, and the measurement-only period, covering days 11–20.

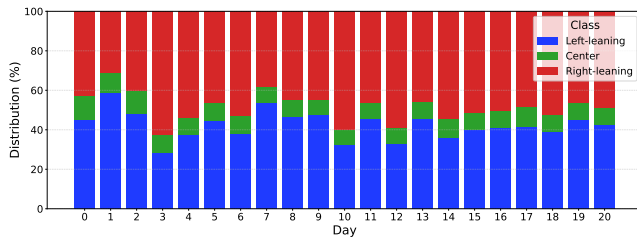
Period	p -value
Days 1–10	3.30×10^{-4}
Days 11–20	1.82×10^{-4}



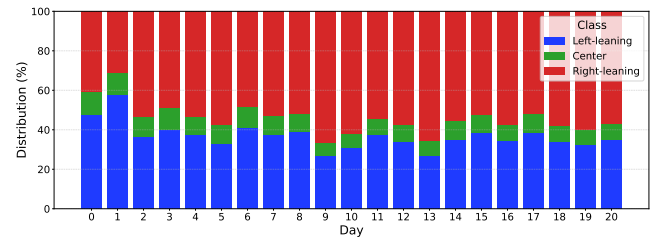
(a) Extreme-left group



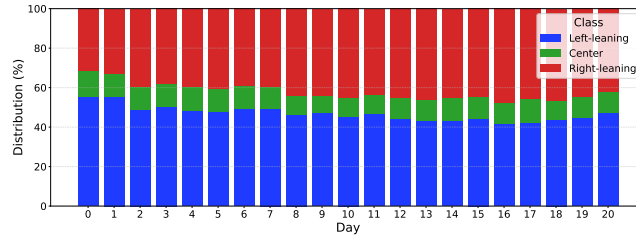
(b) Left group



(c) Right group



(d) Extreme-right group



(e) Control group

Figure 3: Day-by-day evolution of the weighted political-bias distribution in the tracking-permissive environment. Each subfigure corresponds to one news consumption group. The distributions show the share of recommendations classified as left-leaning (*far-left*, *left*), center, or right-leaning (*anti-woke*, *right*, *far-right*) on each day. Day 0 is the baseline collected during the setting phase. Days 1-10 correspond to the period with both news consumption and measurement phases, and days 11-20 correspond to the measurement-only period.